

# EXPLORATORY DATA ANALYSIS OF SAAS PRODUCT USAGE & CUSTOMER RETENTION

Subscription Behaviour, Product  
Usage & Customer Retention

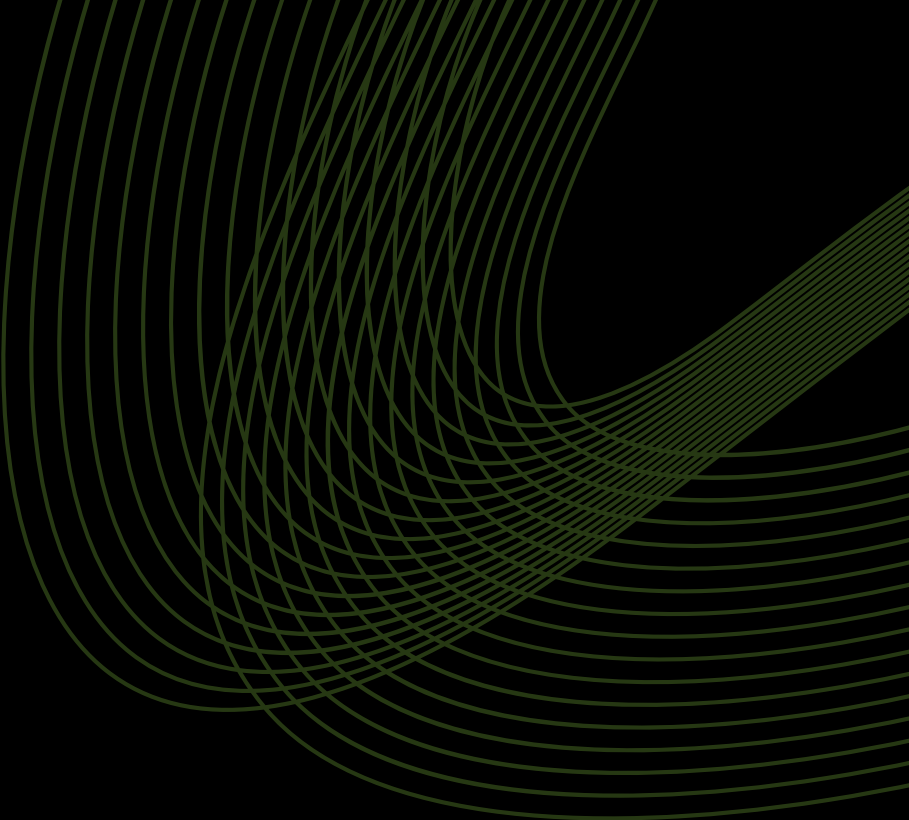


## TOOLS USED

Python, Pandas, Matplotlib, Seaborn & Plotly

## PRESENTED BY

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# TABLE OF CONTENTS

|                               |    |
|-------------------------------|----|
| Problem Statement & Objective | 3  |
| EDA Workflow                  | 4  |
| Key Questions Explored        | 5  |
| Data Overview                 | 6  |
| Insights                      | 13 |
| Business/Developer Takeaways  | 34 |
| Future Work                   | 36 |
| Conclusion                    | 37 |

# PROBLEM STATEMENT & OBJECTIVE

## Problem Statement:

Modern SaaS businesses operate across diverse customer segments with varying usage and retention behaviors, making it challenging to pinpoint the engagement and operational factors that influence churn and long-term revenue.

## Objective:

To explore the SaaS customer usage and retention dataset to uncover patterns, insights, and trends that can help businesses make data-driven decisions around customer engagement, retention, and revenue growth.



# EDA WORKFLOW

For this analysis, a structured workflow was followed, involving data collection, understanding, cleaning, exploration, and summarization of insights to allow a clear understanding of the dataset and its trends.

01

## Data Collection

- Gathered the Google Play Store dataset from Kaggle

02

## Data Understanding & Anomaly Detection

- Looked at data distributions
- Found missing values, outliers, and unusual patterns

03

## Data Cleaning & Treatment

- Fixed missing or incorrect values
- Standardized formats for consistency

04

## Exploratory Analysis

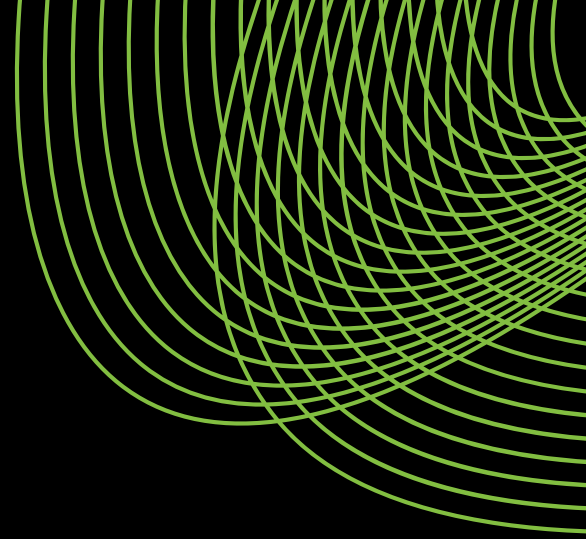
- Univariate Analysis: e.g Customer Segment, Complaint Type, Contract Type etc
- Bivariate Monthly Logins vs Features Used
- Multivariate Analysis: CSAT vs Contract Type vs Customer Segment

05

## Insights & Reporting

- Summarized patterns and trends
- Highlighted key findings for developers and businesses

# KEY QUESTIONS EXPLORED



- How do user demographics (age and gender) influence shopping behavior and engagement patterns on the platform?
- Which stages of the shopping journey (browsing, cart, checkout) contribute most to drop-offs and missed conversions?
- Do occasional shoppers present a greater growth opportunity compared to already frequent buyers?
- How effective are product recommendations in influencing purchase decisions, and where do they fall short?
- What role does pricing perception play in cart abandonment and cross-platform comparison behavior?
- How do customer support responsiveness and delivery reliability impact overall satisfaction and trust?
- How do reviews and post-purchase experiences influence future purchase intent and platform credibility?
- Which satisfaction drivers most strongly contribute to repeat purchases and long-term loyalty?



# DATA OVERVIEW

The dataset provides insights into SaaS customer behavior, covering demographics, subscription characteristics, product usage, billing activity & support interactions. It also captures customer engagement signals, satisfaction metrics, and feedback indicators that are relevant for understanding retention and churn dynamics.

Data Source: Kaggle

## Dataset Size

10000

Records

32

Features

## Purchase Diversity

3

Customer Segments

7

Countries

3

Contract Types



# DATA OVERVIEW

Below is a detailed description of the feature set:

| Dataset Features    | Type                   | Feature Description                                                     |
|---------------------|------------------------|-------------------------------------------------------------------------|
| customer_id         | Date                   | Unique identifier assigned to each customer                             |
| gender              | Categorical            | Gender of the customer                                                  |
| age                 | Numerical              | Age of the customer in years                                            |
| country             | Categorical            | Country where the customer is located                                   |
| customer_segment    | Categorical            | Business-defined customer classification (Individual, SMB, Enterprise)  |
| tenure_months       | Numerical (Discrete)   | Number of months the customer has been subscribed to the service        |
| signup_channel      | Categorical            | Channel through which the customer signed up (Organic, Ads, Referral)   |
| contract_type       | Categorical            | Type of subscription contract (Monthly, Quarterly, Annual)              |
| monthly_logins      | Numerical (Discrete)   | Total number of platform logins by the customer in a month              |
| weekly_active_days  | Numerical (Discrete)   | Average number of days per week the customer actively used the platform |
| avg_session_time    | Numerical (Continuous) | Average duration of a customer session on the platform                  |
| features_used       | Numerical (Discrete)   | Number of distinct product features used by the customer                |
| usage_growth_rate   | Numerical (Continuous) | Percentage change in usage over a recent time period                    |
| last_login_days_ago | Numerical (Discrete)   | Number of days since the customer last logged into the platform         |
| monthly_fee         | Numerical (Discrete)   | Monthly subscription fee charged to the customer                        |
| total_revenue       | Numerical (Discrete)   | Total revenue generated from the customer over their lifetime           |

# DATA OVERVIEW

Continued..

| Dataset Features       | Type                   | Feature Description                                                                  |
|------------------------|------------------------|--------------------------------------------------------------------------------------|
| city                   | Categorical            | City where the customer resides                                                      |
| payment_method         | Categorical            | Payment method used by the customer                                                  |
| payment_failures       | Numerical (Discrete)   | Number of failed payment attempts recorded                                           |
| discount_applied       | Categorical            | Indicates whether a discount was applied to the subscription                         |
| price_increase_last_3m | Categorical            | Indicates whether the customer experienced a price increase in the last three months |
| support_tickets        | Numerical (Discrete)   | Number of customer support tickets raised                                            |
| avg_resolution_time    | Numerical (Continuous) | Average time taken to resolve customer support tickets                               |
| complaint_type         | Categorical            | Category of complaint raised by the customer                                         |
| csat_score             | Numerical (Continuous) | Customer Satisfaction Score                                                          |
| escalations            | Numerical (Discrete)   | Number of support cases escalated beyond first-level support                         |
| email_open_rate        | Numerical (Continuous) | Percentage of marketing or communication emails opened                               |
| marketing_click_rate   | Numerical (Continuous) | Percentage of marketing emails where links were clicked                              |
| nps_score              | Numerical (Discrete)   | Net Promoter Score provided by the customer                                          |
| survey_response        | Categorical            | Indicates whether the customer responded to surveys                                  |
| referral_count         | Numerical (Discrete)   | Number of new customers referred by the existing customer                            |
| churn                  | Numerical (Binary)     | Indicates whether the customer discontinued the service (0 = No, 1 = Yes)            |